DATABASE SYSTEMS | LAB MANUAL

Database Normalization

Complete Summary Report

Hospital Patient Visits | Assessment Problem — Task 8

XAMPP / MySQL 8.x | Course: Database Systems

UNF Raw Data → **1NF** Atomic → **2NF** No Partial Deps → **3NF** No Trans. Deps

WHAT'S INSIDE

✦ Functional Dependencies & Anomaly Analysis

✦ Step-by-Step UNF → 1NF → 2NF → 3NF Transformation

✦ Before & After Tables for Each Stage

✦ Complete MySQL Schema & SQL Code

✦ Anomaly Elimination Proof

1. The Problem — Unnormalized Data (UNF)

The hospital stores all consultation data in one single flat table. Every row repeats patient info, doctor info, and department info — creating massive redundancy and making the database fragile. Before normalization, the table looks like this:

VisitID	VisitDate	PatientID	PatName	PatPhone	DocID	DocName	Specialty	DeptName	DeptHead	Diagnosis	Fee
V-9001	2026-04-10	P-201	Hassan	0300-1112233	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Hypertension	2500
V-9002	2026-04-10	P-202	Mehreen	0301-4445566	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Eczema	2000
V-9003	2026-04-11	P-201	Hassan	0300-1112233	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Allergy	2000
V-9004	2026-04-12	P-203	Junaid	0302-7778899	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Arrhythmia	3000

■ Red = Redundant data repeated across rows | Hassan (P-201) appears twice | Dr. Imran & Dr. Asma info repeated

Anomalies in UNF

Anomaly Type	Problem	Concrete Example
Insertion	Cannot add a new doctor until a patient visits them.	Dr. Zara (Neurology) cannot be recorded without a patient visit.
Update	Changing one fact requires updating multiple rows.	Dr. Imran moves office → must update rows V-9001 AND V-9004.
Deletion	Deleting a visit accidentally removes other facts.	Delete V-9002 → lose all info about Dr. Asma if no other visits.

Identified Functional Dependencies

Functional Dependency	Attributes Determined	Note
VisitID →	VisitDate, Diagnosis, Fee, PatientID, DoctorID	Direct — VisitID is the PK
PatientID →	PatientName, PatientPhone	Patient info per patient
DoctorID →	DoctorName, Specialty, DeptName	Doctor info per doctor
DeptName →	DeptHead	Transitive: dept determines head

2. First Normal Form (1NF) — Atomic Values & Primary Key

Rule: Every cell holds a single atomic value | No repeating groups | Primary Key must be defined

The UNF table is converted to 1NF by declaring **VisitID** as the Primary Key and ensuring every cell contains exactly one value. The data remains in one table — 1NF does NOT split tables. This step fixes atomicity only; redundancy still exists and will be fixed in 2NF.

UNF → 1NF Changes	• No multi-valued cells → all cells already atomic in this dataset ■ • Primary Key declared: **VisitID** • Number of tables: still 1 (HospitalVisit_1NF) • Number of columns: 12 | Number of rows: 4

■ 1NF ACHIEVED — 1 table | PK = VisitID | Atomic values | 12 columns | Redundancy still present

sitID (PK)*	VisitDate	PatientID	PatientName	PatientPhone	DoctorID	DoctorName	Specialty	DeptName	DeptHead	Diagnosis	Fe
V-9001	2026-04-10	P-201	Hassan	0300-1112233	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Hypertension	250
V-9002	2026-04-10	P-202	Mehreen	0301-4445566	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Eczema	200
V-9003	2026-04-11	P-201	Hassan	0300-1112233	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Allergy	200
V-9004	2026-04-12	P-203	Junaid	0302-7778899	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Arrhythmia	300

3. Second Normal Form (2NF) — Remove Partial Dependencies

Rule: Must be in 1NF + Every non-prime attribute depends on the WHOLE primary key (no partial deps)

PatientName & PatientPhone depend on PatientID alone — not on VisitID. DoctorName, Specialty, DeptName & DeptHead depend on DoctorID alone. These are partial dependencies. Fix: split into 3 tables.

Patient

PatientID (PK)	PatientName	PatientPhone
P-201	Hassan	0300-1112233
P-202	Mehreen	0301-4445566
P-203	Junaid	0302-7778899

Doctor

DoctorID (PK)	DoctorName	Specialty	DeptName	DeptHead ■■
D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq
D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma

Visit

VisitID (PK)	VisitDate	PatientID (FK)	DoctorID (FK)	Diagnosis	Fee
V-9001	2026-04-10	P-201	D-30	Hypertension	2500
V-9002	2026-04-10	P-202	D-31	Eczema	2000
V-9003	2026-04-11	P-201	D-31	Allergy	2000
V-9004	2026-04-12	P-203	D-30	Arrhythmia	3000

■■ Doctor table still has DeptName → DeptHead (transitive dependency) — fixed in 3NF

■ 2NF ACHIEVED — 3 tables | Partial deps removed | Patient & Doctor data stored once | FKs defined

4. Third Normal Form (3NF) — Remove Transitive Dependencies

Rule: Must be in 2NF + No non-prime attribute depends on another non-prime attribute

In the Doctor table: **DoctorID → DeptName → DeptHead**. DeptHead depends on DeptName (a non-key attribute), not directly on DoctorID. Fix: Extract Department table. DeptName becomes a FK in Doctor.

DoctorID (PK) →■ **DeptName (non-key)** → **DeptHead (non-key)** **Transitive! Must be extracted**

Patient (unchanged)

PatientID (PK)	PatientName	PatientPhone

P-201	Hassan	0300-1112233
P-202	Mehreen	0301-4445566
P-203	Junaid	0302-7778899

Department (NEW ■)

DeptName (PK)	DeptHead
Heart Care	Dr. Tariq
Skin Clinic	Dr. Asma

Doctor (DeptHead removed)

DoctorID (PK)	DoctorName	Specialty	DeptName (FK)
D-30	Dr. Imran	Cardiology	Heart Care
D-31	Dr. Asma	Dermatology	Skin Clinic

Visit (unchanged)

VisitID (PK)	VisitDate	PatientID (FK)	DoctorID (FK)	Diagnosis	Fee
V-9001	2026-04-10	P-201	D-30	Hypertension	2500
V-9002	2026-04-10	P-202	D-31	Eczema	2000
V-9003	2026-04-11	P-201	D-31	Allergy	2000
V-9004	2026-04-12	P-203	D-30	Arrhythmia	3000

■ 3NF ACHIEVED — 4 tables | All transitive deps removed | Every fact stored exactly once | ALL anomalies eliminated

5. Master Comparison Table — UNF → 1NF → 2NF → 3NF

Feature	UNF	1NF	2NF	3NF
Number of Tables	1	1	3	4
Primary Key	None	VisitID	VisitID, PatientID, DoctorID	VisitID, PatientID, DoctorID, DeptName
Foreign Keys	None	None	Visit→Patient, Visit→Doctor	+ Doctor→Department
Atomic Values	■	■	■	■
Repeating Groups	■	■	■	■
Partial Deps	■	■	**■ Removed**	**■ None**
Transitive Deps	■	■	**■■ DeptName→DeptHead**	**■ Removed**
Insertion Anomaly	■	■	**■ Patient/Doctor OK**	**■ All entities independent**
Update Anomaly	■	■	**■ Mostly fixed**	**■ Fully fixed**
Deletion Anomaly	■	■	**■■ Dept still at risk**	**■ Fully fixed**
Redundancy Level	Very High	High	Low	None
Data Integrity	Poor	Basic	Good	Excellent

6. Final 3NF Schema Relationships

Patient
■■■■■■■■■
PatientID (PK)
PatientName
PatientPhone

→→

Visit
■■■■■■■■■
VisitID (PK)
VisitDate
PatientID (FK)
DoctorID (FK)
Diagnosis Fee

Doctor
■■■■■■■■■
DoctorID (PK)
DoctorName
Specialty
DeptName (FK)

Department
■■■■■■■■■
DeptName
(PK)
DeptHead

PatientIDDoctorIDDeptName

7. Complete MySQL Code (3NF — Final Schema)

```
USE hospital_normalization; -- Patient table CREATE TABLE Patient ( PatientID VARCHAR(10)
PRIMARY KEY, PatientName VARCHAR(50) NOT NULL, PatientPhone VARCHAR(20) NOT NULL ); --
Department table (extracted from Doctor) CREATE TABLE Department ( DeptName VARCHAR(50)
PRIMARY KEY, DeptHead VARCHAR(50) NOT NULL ); -- Doctor table (DeptName is FK →
Department) CREATE TABLE Doctor ( DoctorID VARCHAR(10) PRIMARY KEY, DoctorName VARCHAR(50)
NOT NULL, Specialty VARCHAR(50) NOT NULL, DeptName VARCHAR(50) NOT NULL, FOREIGN KEY
(DeptName) REFERENCES Department(DeptName) ); -- Visit table (FK → Patient, Doctor)
CREATE TABLE Visit ( VisitID VARCHAR(10) PRIMARY KEY, VisitDate DATE NOT NULL, PatientID
VARCHAR(10) NOT NULL, DoctorID VARCHAR(10) NOT NULL, Diagnosis VARCHAR(100) NOT NULL, Fee
INT NOT NULL, FOREIGN KEY (PatientID) REFERENCES Patient(PatientID), FOREIGN KEY
```

```
(DoctorID) REFERENCES Doctor(DoctorID) ); -- Rebuild original report (4-table JOIN) SELECT
v.VisitID, v.VisitDate, p.PatientName, p.PatientPhone, d.DoctorName, d.Specialty,
d.DeptName, dept.DeptHead, v.Diagnosis, v.Fee FROM Visit v JOIN Patient p ON v.PatientID =
p.PatientID JOIN Doctor d ON v.DoctorID = d.DoctorID JOIN Department dept ON d.DeptName =
dept.DeptName ORDER BY v.VisitDate;
```

8. Key Takeaways

1NF	Ensures atomic values and defines a Primary Key. Foundation of all normalization.
2NF	Removes partial dependencies. Each non-key attribute depends on the full PK. Tables split by entity.
3NF	Removes transitive dependencies. No non-key column determines another non-key column.
Goal	Every fact stored in exactly one place. Data redundancy = 0. All anomalies = eliminated.

■ Final Result: 1 unnormalized table with 12 columns → 4 clean normalized tables | Patient Visit → Doctor → Department | Zero redundancy | Full referential integrity